MSG No. 201.21 $P_In\bar{3}$ [Type IV, cubic]

Table 1: Wyckoff site: 2a, site symmetry: m'-3'.

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,37,38,39,40,41,42,43,44,45,46,47,48]
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36]

Table 2: Wyckoff site: 6b, site symmetry: m'm'm'..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,2,3,4,37,38,39,40]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[5,8,10,12,41,44,46,48]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[6,7,9,11,42,43,45,47]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[13,14,15,16,25,26,27,28]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[17,20,22,24,29,32,34,36]
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[18,19,21,23,30,31,33,35]

Table 3: Wyckoff site: 8c, site symmetry: .-3.

No.	position	mapping
1	$[0, 0, 0]$	[1,5,6,13,17,18]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,10,11,14,22,23]
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[3,7,12,15,19,24]
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[4,8,9,16,20,21]
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[25,29,30,37,41,42]
6	$\left[\frac{1}{2}, 0, 0\right]$	[26,34,35,38,46,47]
7	$\left[0, \frac{1}{2}, 0\right]$	[27,31,36,39,43,48]
8	$\left[0, 0, \frac{1}{2}\right]$	[28,32,33,40,44,45]

Table 4: Wyckoff site: 12d, site symmetry: 2m'm'..

No.	position	mapping
1	$\left[x, \frac{3}{4}, \frac{3}{4}\right]$	[1,2,39,40]
2	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	[3,4,37,38]
3	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	[5,12,44,46]
4	$\left[\frac{3}{4}, \frac{3}{4}, x\right]$	[6,9,43,47]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[7,11,42,45]
6	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[8,10,41,48]
7	$\left[-x, \frac{1}{4}, \frac{1}{4}\right]$	[13,14,27,28]
8	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	[15,16,25,26]
9	$\left[\frac{1}{4}, -x, \frac{1}{4}\right]$	[17,24,32,34]
10	$\left[\frac{1}{4}, \frac{1}{4}, -x\right]$	[18,21,31,35]
11	$\left[\frac{1}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[19,23,30,33]
12	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[20,22,29,36]

Table 5: Wyckoff site: `12e`, site symmetry: `2m'm'..`

No.	position	mapping
1	$\left[x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,2,39,40]`
2	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,4,37,38]`
3	$\left[\frac{1}{4},\ x,\ \frac{3}{4}\right]$	`[5,12,44,46]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ x\right]$	`[6,9,43,47]`
5	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-x\right]$	`[7,11,42,45]`
6	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	`[8,10,41,48]`
7	$\left[-x,\ \frac{1}{4},\ \frac{3}{4}\right]$	`[13,14,27,28]`
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[15,16,25,26]`
9	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	`[17,24,32,34]`
10	$\left[\frac{1}{4},\ \frac{3}{4},\ -x\right]$	`[18,21,31,35]`
11	$\left[\frac{1}{4},\ \frac{3}{4},\ x+\frac{1}{2}\right]$	`[19,23,30,33]`
12	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	`[20,22,29,36]`

Table 6: Wyckoff site: `16f`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,6]`
2	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[2,10,11]`
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}-x\right]$	`[3,7,12]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ x\right]$	`[4,8,9]`
5	$[-x,\ -x,\ -x]$	`[13,17,18]`
6	$\left[-x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[14,22,23]`
7	$\left[x+\frac{1}{2},\ -x,\ x+\frac{1}{2}\right]$	`[15,19,24]`
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -x\right]$	`[16,20,21]`
9	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[25,29,30]`
10	$\left[x+\frac{1}{2},\ -x,\ -x\right]$	`[26,34,35]`
11	$\left[-x,\ x+\frac{1}{2},\ -x\right]$	`[27,31,36]`
12	$\left[-x,\ -x,\ x+\frac{1}{2}\right]$	`[28,32,33]`
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[37,41,42]`
14	$\left[\frac{1}{2}-x,\ x,\ x\right]$	`[38,46,47]`
15	$\left[x,\ \frac{1}{2}-x,\ x\right]$	`[39,43,48]`
16	$\left[x,\ x,\ \frac{1}{2}-x\right]$	`[40,44,45]`

Table 7: Wyckoff site: `24g`, site symmetry: `m'..`

No.	position	mapping
1	$\left[\frac{3}{4},\ y,\ z\right]$	`[1,38]`
2	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[2,37]`
3	$\left[\frac{3}{4},\ y,\ \frac{1}{2}-z\right]$	`[3,40]`
4	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z\right]$	`[4,39]`
5	$\left[z,\ \frac{3}{4},\ y\right]$	`[5,48]`
6	$\left[y,\ z,\ \frac{3}{4}\right]$	`[6,45]`
7	$\left[\frac{1}{2}-y,\ z,\ \frac{3}{4}\right]$	`[7,47]`
8	$\left[\frac{1}{2}-z,\ \frac{3}{4},\ y\right]$	`[8,46]`

continued ...

Table 7

No.	position	mapping
9	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ \frac{3}{4}\right]$	[9,42]
10	$\left[z,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[10,44]
11	$\left[y,\ \frac{1}{2}-z,\ \frac{3}{4}\right]$	[11,43]
12	$\left[\frac{1}{2}-z,\ \frac{3}{4},\ \frac{1}{2}-y\right]$	[12,41]
13	$\left[\frac{1}{4},\ -y,\ -z\right]$	[13,26]
14	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[14,25]
15	$\left[\frac{1}{4},\ -y,\ z+\frac{1}{2}\right]$	[15,28]
16	$\left[\frac{1}{4},\ y+\frac{1}{2},\ -z\right]$	[16,27]
17	$\left[-z,\ \frac{1}{4},\ -y\right]$	[17,36]
18	$\left[-y,\ -z,\ \frac{1}{4}\right]$	[18,33]
19	$\left[y+\frac{1}{2},\ -z,\ \frac{1}{4}\right]$	[19,35]
20	$\left[z+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[20,34]
21	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{4}\right]$	[21,30]
22	$\left[-z,\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[22,32]
23	$\left[-y,\ z+\frac{1}{2},\ \frac{1}{4}\right]$	[23,31]
24	$\left[z+\frac{1}{2},\ \frac{1}{4},\ y+\frac{1}{2}\right]$	[24,29]

Table 8: Wyckoff site: 48h, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[2]
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[4]
5	$[z,\ x,\ y]$	[5]
6	$[y,\ z,\ x]$	[6]
7	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[7]
8	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[8]
9	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[9]
10	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[10]
11	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[11]
12	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[16]
17	$[-z,\ -x,\ -y]$	[17]
18	$[-y,\ -z,\ -x]$	[18]
19	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[19]
20	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	[20]
21	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[21]
22	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[22]
23	$\left[-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[23]
24	$\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$	[24]
25	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[25]

continued ...

Table 8

No.	position	mapping
26	$\left[x+\frac{1}{2},\,-y,\,-z\right]$	[26]
27	$\left[-x,\,y+\frac{1}{2},\,-z\right]$	[27]
28	$\left[-x,\,-y,\,z+\frac{1}{2}\right]$	[28]
29	$\left[z+\frac{1}{2},\,x+\frac{1}{2},\,y+\frac{1}{2}\right]$	[29]
30	$\left[y+\frac{1}{2},\,z+\frac{1}{2},\,x+\frac{1}{2}\right]$	[30]
31	$\left[-y,\,z+\frac{1}{2},\,-x\right]$	[31]
32	$\left[-z,\,-x,\,y+\frac{1}{2}\right]$	[32]
33	$\left[-y,\,-z,\,x+\frac{1}{2}\right]$	[33]
34	$\left[z+\frac{1}{2},\,-x,\,-y\right]$	[34]
35	$\left[y+\frac{1}{2},\,-z,\,-x\right]$	[35]
36	$\left[-z,\,x+\frac{1}{2},\,-y\right]$	[36]
37	$\left[\frac{1}{2}-x,\,\frac{1}{2}-y,\,\frac{1}{2}-z\right]$	[37]
38	$\left[\frac{1}{2}-x,\,y,\,z\right]$	[38]
39	$\left[x,\,\frac{1}{2}-y,\,z\right]$	[39]
40	$\left[x,\,y,\,\frac{1}{2}-z\right]$	[40]
41	$\left[\frac{1}{2}-z,\,\frac{1}{2}-x,\,\frac{1}{2}-y\right]$	[41]
42	$\left[\frac{1}{2}-y,\,\frac{1}{2}-z,\,\frac{1}{2}-x\right]$	[42]
43	$\left[y,\,\frac{1}{2}-z,\,x\right]$	[43]
44	$\left[z,\,x,\,\frac{1}{2}-y\right]$	[44]
45	$\left[y,\,z,\,\frac{1}{2}-x\right]$	[45]
46	$\left[\frac{1}{2}-z,\,x,\,y\right]$	[46]
47	$\left[\frac{1}{2}-y,\,z,\,x\right]$	[47]
48	$\left[z,\,\frac{1}{2}-x,\,y\right]$	[48]